T-T-871E
June 18, 1970
SUPERSEDING
Fed. Spec. T-T-871d
November 24, 1965

FEDERAL SPECIFICATION

TWINE, COTTON, WRAPPING

This specification was approved by the commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements of cotton wrapping twine (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The twine shall be of the following types and classes:

Type I - Natural.
Type II - Mildew-resistant.

Class 1 - Copper-8-quinolinolate.
Class 2 - 2,2'-methylenebis-(4-chlorophenol).

1.2.2 Plies. The twine shall be of the following plies, as specified (see table I and 6.2):

Ply		
3	8	24
4	12	A
5	16	B
6	20	-

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-636 - Box, Fiberboard.

FSC 4020

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 191 - Textile, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-3131 - Cordage, Preparation for Delivery Of.
MIL-T-3530 - Thread and Twine; Mildew Resistant or Water Repellent Treated.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Government and suppliers' purchase. The requirements specified in 3.8 apply only to twine purchased directly by the Government. All other requirements apply to twine purchased by a supplier as a component for an end item and to twine purchased directly by the Government.

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3.2 Material. The twine shall be constructed from cotton of suitable grade and staple to meet the requirements of this specification.

3.3 Construction and physical requirements. The twine shall be constructed from three or more twisted single yarns plied as specified, except A and B ply. The twine shall be "Z" twist and shall conform to the requirements as specified in table I when tested as specified in 4.2.5.

TABLE I. Physical requirements

Ply (size)	Length per pound, feet (minimum)	Breaking strength, pounds (minimum)
3	6000	5.5
4	4500	8
5	3600	10
6	3000	12
8	2250	16
12	1500	24
16	1125	32
20	900	40
24	750	48
A 1/	950	40
B 1/	1950	20.5

1/ Ply is not standard with manufacturers. In no instance shall twine be less than 2 ply.

3.4 Color. Unless otherwise specified, the twine shall be furnished in the natural color (see 6.2 and 6.4). When mildew-resistant twine is specified, the color shall be that shade imparted by the treatment on natural-colored twine.

3.5 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material for untreated twine shall not exceed 5.0 percent when tested as specified in 4.2.5.

3.6 Finish.

3.6.1 Type I. Type I twine shall have a natural finish and shall conform to all the requirements specified herein. Natural finish shall denote a twine for which no consistency or luster, other than that inherent with the cotton material, is required.

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3.6.2 Type II. Type II twine shall conform to the requirements specified for type I twine, and in addition shall be mildew-resistant treated as specified for the appropriate class.

3.6.2.1 Type II, class 1. Class 1 twine shall be mildew-resistant treated with copper-8-quinolinolate in accordance with type I, class 1 treatment of MIL-T-3530.

3.6.2.2 Type II, class 2. Class 2 twine shall be mildew-resistant treated with 2,2'-methylenebis-(4-chlorophenol) in accordance with type I, class 2 treatment of MIL-T-3530.

3.7 Elongation (type I, A and B ply twine only). The elongation of the type I, A and B ply twines shall not exceed 12 percent when tested as specified in 4.2.5.

3.8 Put up. Except as required in 3.8.1, and unless otherwise specified, type I and II twine shall be furnished in 1/2, 1, 2, 2-1/2, or 3 pound net weight balls, tubes or cones as specified (see 6.2). A plus or minus tolerance of 10 percent in weight shall be allowed on each holder provided the average net weight does not fall below the net weight specified above for the applicable holder when examined as specified in 4.2.3. Each holder of twine shall contain not more than the applicable maximum allowable number of knots and splices specified in table II. The twine shall be continuous throughout and shall be so wound that each turn and layer is free from entanglement.

3.8.1 Type I, A and B ply twines only. Twines of A and B ply construction shall be furnished on cones weighing 5 to 6-1/2 pounds inclusive, gross weight. The base of the put-up cone shall measure not more than 6-7/8 inches in diameter. Each cone of twine shall contain not more than the applicable maximum allowable number of knots and splices specified in table II. The twine shall be continuous throughout and shall be so wound that each turn and layer is free from entaglement. Each cone shall have a red colored trailer or marker approximately 20 feet long attached [illegible] of the inside end of the twine.

TABLE II. Maximum allowable number of knots and splices 1/

Put-up ball, tube, or cone	Maximum number of knots and splices per holder
1/2 pound holder	1
1 pound holder	2
2 and 2-1/2 pound holder	3

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TABLE II. Maximum allowable number of knots and splices 1/ (cont'd)

Put-up ball, tube, or cone	Maximum number of knots and splices per holder
3-pound holder	4
5 to 6-1/2 pound cone (A and B ply twine)	1

1/ A knot shall be considered to exist when the strands of the twine are separated and tied in one place joining two distinct pieces of twine. The term "knot" shall not include the tying of knots joining individual strands of twine.

3.9 Workmanship. The twine shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance (DOD procurements only). Where certificates of compliance are submitted, the Government reserves the right to check test [illegible] to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item for visual defects. The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one ball, tube or cone. Ten

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percent of the length contained in each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of balls, tubes or cones. The acceptable quality level shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

TABLE III. Visual examination

Examine	Defect	Classification	
		Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
	Kink, broken or loose end	X	
Color	Other than natural (unless otherwise specified)	X	
Cleanness	Spot or stain - clearly visible 1/		X
	Lump, picker waste or similar dirty waste, and linters clearly visible		X
Finish	Other than specified	X	

1/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for weight and winding. The sample unit for this examination shall be one ball, tube or cone. The inspection level shall be S-3 and the acceptable quality level shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size (number of sample units) shall be 10 and the acceptance number 1. The lot size shall be the number of balls, tubes or cones in the inspection lot. Defects shall be as listed in 4.2.3.1 and 4.2.3.2.

4.2.3.1 Examination for weight.

4.2.3.1.1 Net weight per ball, tube or cone. A defect with regard to weight shall be considered to exist if the net weight on any holder is less than the minimum or more than the maximum specified in 3.8.

4.2.3.1.1.1 Gross weight per cone (A and B ply). A defect with regard to weight shall be considered to exist if the gross weight of any one cone is less than the minimum or more than the maximum weights required in 3.8.1.

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4.2.3.1.2 Average weight. No AQL is applicable to this examination for average weight. A lot shall be unacceptable if the average net weight is less than the applicable weight specified in 3.8 or the average gross weight is less than the minimum or more than the maximum weight specified in 3.8.1

4.2.3.2 Examination for winding. Defects with regard to winding shall be considered to exist if any of the following are determined during the examination:

(a) Improperly or not firmly wound resulting in kinks, entangling, or slippage during unwinding or otherwise affecting free unhampered unwinding of twine.

(b) Put-up not as specified.

(c) Any holder containing more than the maximum allowable number of knots and splices specified for the applicable put-up (see table II).

(d) Not in a continuous length.

4.2.4 Examination of preparation for delivery requirements. An examination shall be made to determine whether the packaging, packing and markings complies with the section 5 requirements of this specification. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

4.2.4.1 Examination for palletization. An examination shall be made to determine whether the palletization complies with the section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimensions	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not banded with required straps as specified.

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Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.2.5 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable and as listed in table V shall be followed. When data in the "Number of determinations" and "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table IV. The sample unit for testing shall be one ball, tube or cone of twine. The lot size shall be expressed in units of one ball, tube or cone. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements including quantity of delivery may be made under prevailing atmospheric conditions, except in settlement of dispute in which case the test shall be made upon material which has reached equilibrium under standard conditions as defined in section 4 of Fed. Std. No. 191.

TABLE IV. Sampling for tests

Lot size (balls, tubes, or cones)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V. Test methods

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as:
Material identification	3.2	1200 1/	1	Pass or fail
Direction of twist	3.3	4050	1	Pass or fail

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TABLE V. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as:
Ply (size)	Table I	Visual	1	Pass or fail
Length per pound	Table I	6004	2	-
Breaking strength	Table I	4102	5	-
Elongation (A and B ply twine only)	3.7	4102	5	-
Nonfibrous material	3.5	2611	-	-

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A and C. Twine, put up as specified in 3.8 shall be packaged in accordance with applicable requirements of MIL-C-3131.

5.1.2 Level B.

5.1.2.1 Standard pack (Civil agencies). Twine, put up as specified in 3.8, of one type, class and ply (size) shall be in ball or tube form or wound in a cone in accordance with commercial practice. Cones shall be wrapped in 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268 and secured with adhesive, gummed tape, pressure-sensitive tape or tucked inside the base of the cones. Balls require no wrapping (see 6.3).

5.1.2.1.1 A and B ply twine. In addition to the applicable requirement in 5.1.2.1, twine shall be wound on cones having an inside diameter of between 2-1/8 and 3 inches. The height of the finished cone shall not exceed 9-1/2 inches.

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5.1.2.2 Military requirements. Twine shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Unless otherwise specified by civil agencies, twine shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.2.1.1 Standard pack (civil agencies). The number of balls or cones specified in table VI shall be packed in a new shipping container conforming to the requirements of PPP-B-636, style CSSC, FOL, or SFF, type CF, class domestic, variety SW, grade 350 or variety DW, grade 275; or type SF, class domestic, grade 350. Closure shall be in accordance with the appendix of the specification (see 6.3).

TABLE VI. Quantity of balls or cones per shipping container

Weight of ball or cone	Number of balls or cones
1/2 pound ball	120 balls
1 pound ball	60 balls
2 pound cone	30 cones
2 1/2-pound cone	25 cones
3 pound cone	20 cones
5 to 6-1/2-pound cone (A and B ply twine)	12 cones 1/

1/ Not to exceed 80 pounds gross weight.

5.3 Palletization. When specified (see 6.2), twine packed as specified in 5.2.1, shall be palletized in accordance with the applicable requirements of MIL-C-3131. The load type shall be as specified by the procuring activity (see 6.2).

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123. Each unit cone shall be marked or labeled on one side of the wrapping with the applicable Federal Stock Number, specification symbol, the type, class, and ply (size), and the weight of the cone.

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5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. The twine covered by this specification is intended for general purpose use. Type I, A and B ply twines are primarily intended for use by the Post Office Department for one-wrap and two-wrap tying of mail by machine.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Ply (size) required (see 1.2.2 and table I).
(d) Color of twine if other than specified (see 3.4).
(e) Put-up required (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) When palletization is required (see 5.3).
(h) Selection of applicable load types for palletization when required (see 5.3).

6.3 The level B standard pack requirements in 5.1.2.1 and 5.2.1.1 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard packing requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage and subsequent issue or shipment to eventual user.

6.4 Dyed twine. Should dyed twine be required, the colorfastness requirements shall be as specified by the procuring activity.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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